Hybrid Market-Making — Performance Summary

Comparison of Avellaneda–Stoikov baselines, Model-Free PPO, and Hybrid AS+RL (proposed method).

Method	Mean PnL	Delta PnL(vs Static)	Sharpe	Delta Sharpe(vs Static)	Max DD	CVaR 5%	Mean |Inv|	Fill Rate
Constrained AS (Guéant)	63.7435	+0.0826	7.054	+0.012	1.3483	0.3369	1.085	0.2180
Static AS	63.6609	+0.0000	7.042	+0.000	1.3503	0.3385	1.085	0.2180
Hybrid Tuning RL	49.3807	-14.2802	5.006	-2.036	0.9725	0.2176	0.647	0.1180
Hybrid AS+RL (Proposed)	66.2980	+2.6371	4.452	-2.590	4.8815	1.1505	4.076	0.2231
Model-Free RL (PPO)	54.4643	-9.1966	4.557	-2.485	2.8021	0.6312	2.442	0.1246

Table 1. Composite score = weighted z-score across PnL, Sharpe, drawdown, inventory risk, and fill rate. Green row = top-ranked method.

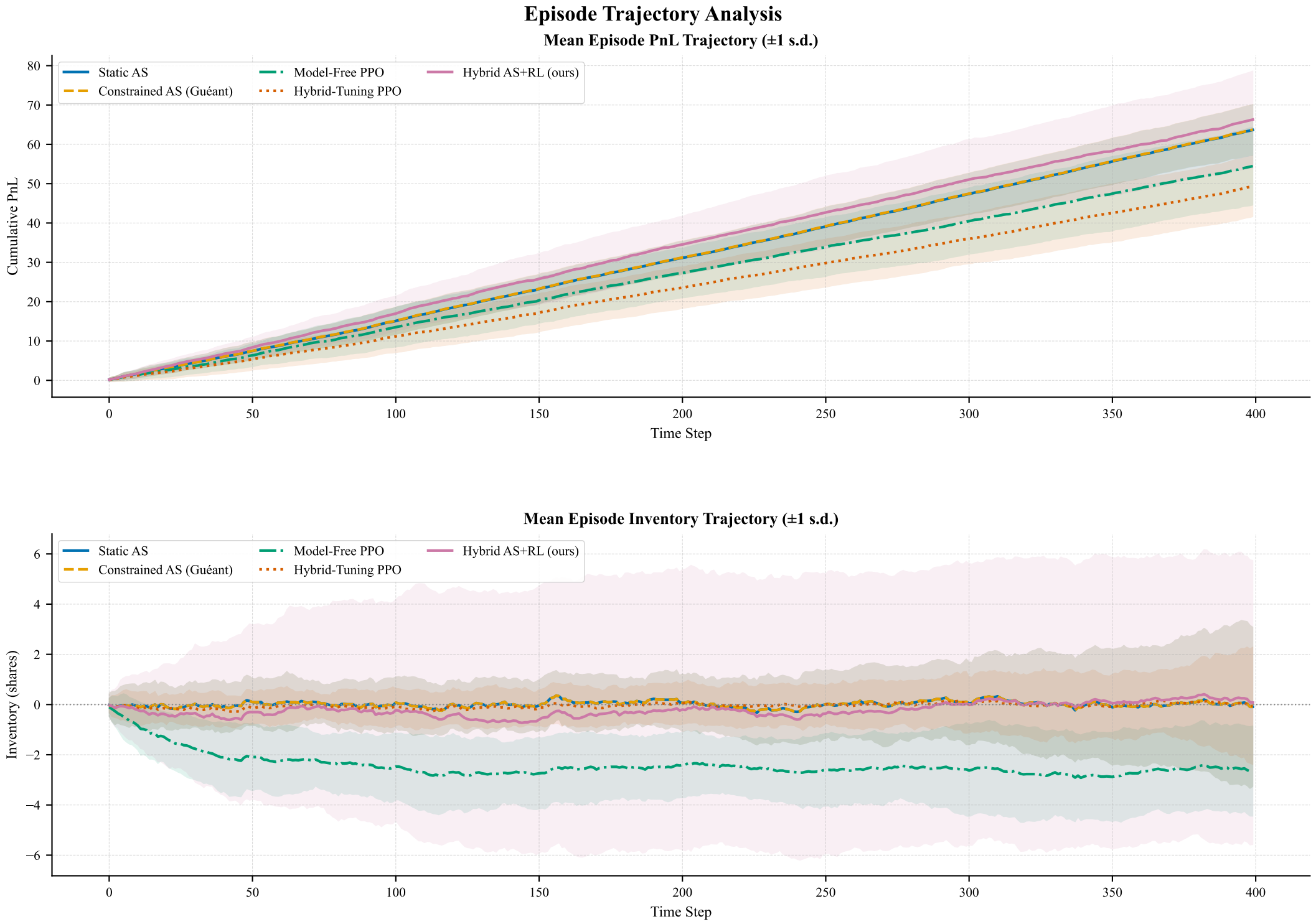

Figure 1. Shaded bands denote ±1 s.d. across evaluation episodes. Inventory converges faster for the constrained and hybrid methods.

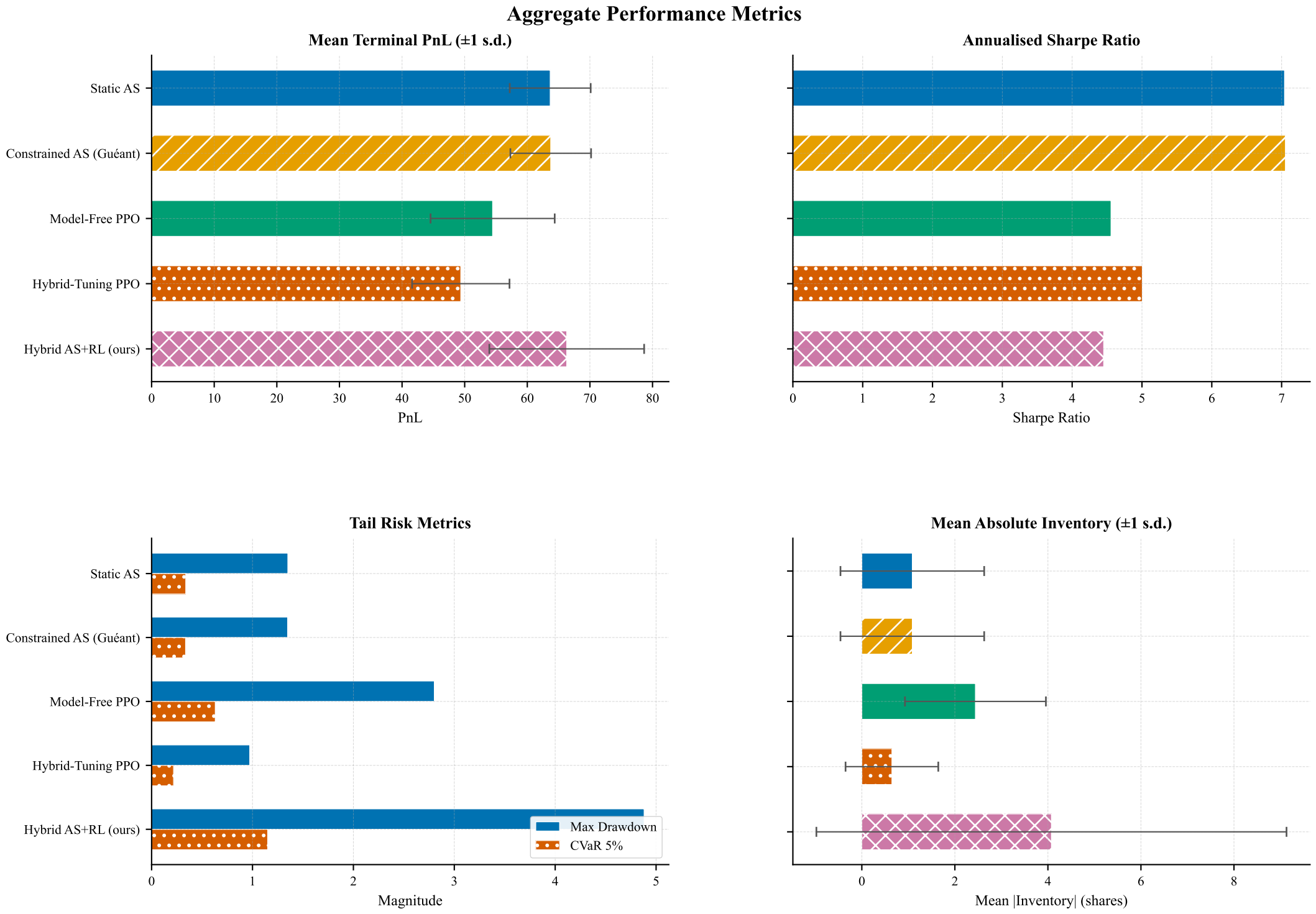

Figure 2. Error bars denote ±1 s.d. across episodes. Hatches distinguish methods; colours follow the Okabe–Ito palette.

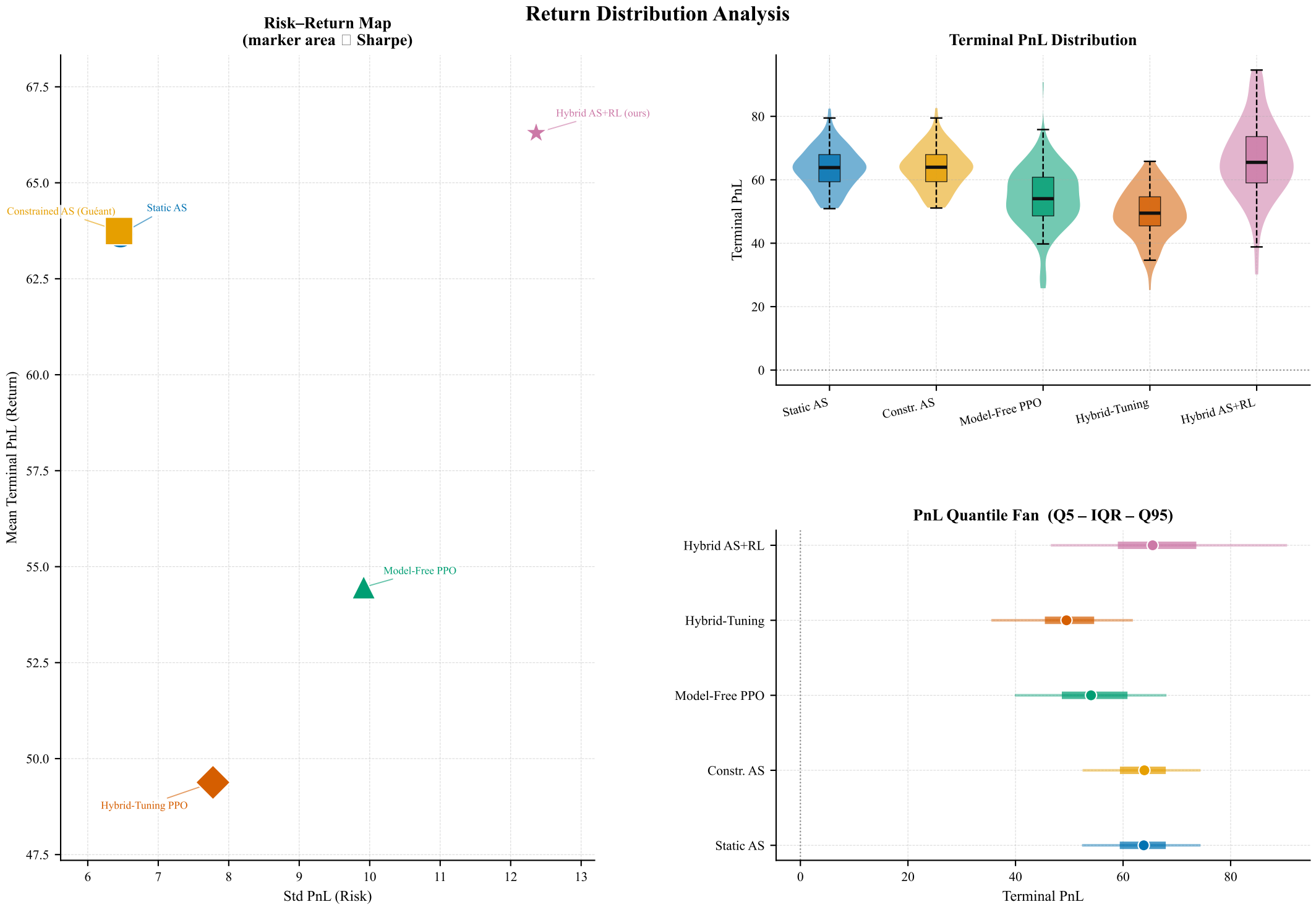

Figure 3. Left: risk–return frontier; marker area □ Sharpe ratio. Top-right: violin + IQR box overlay. Bottom-right: quantile fan (thick bar = IQR, thin lines = Q5–Q95, dot = median).

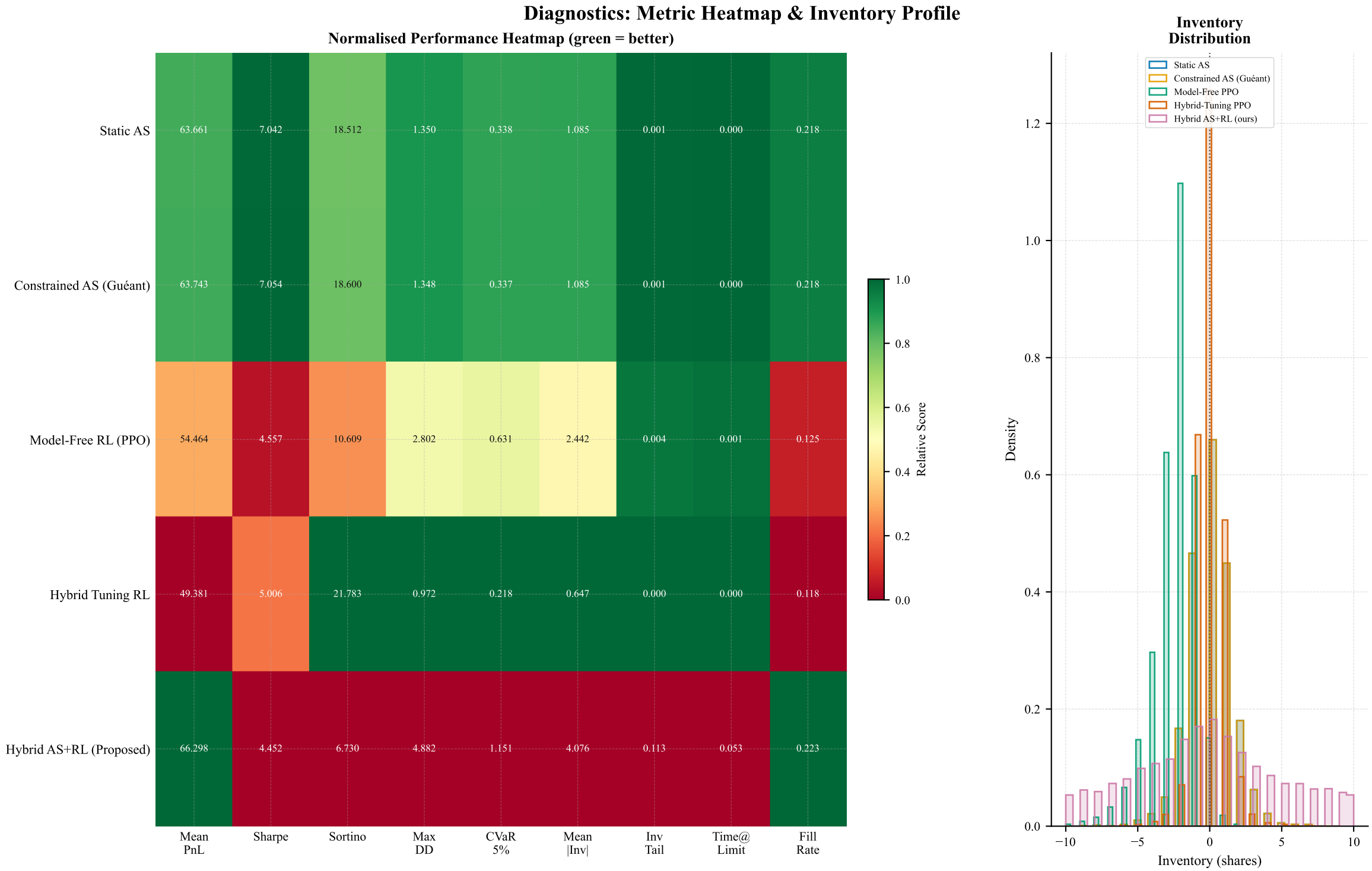

Figure 4. Left: each cell shows the raw metric value; colour normalised column-wise (green = best in column). Right: overlaid inventory density histograms across all evaluation episodes.

Statistical Analysis

Table A. Statistical Significance (Welch t-test vs. Static AS baseline)

Method	N episodes	Mean PnL	Delta vs Static	t-statistic	p-value	Significance	Cohen's d
Static AS	100	63.6609	+0.0000	—	—	—	0.000
Constrained AS (Guéant)	100	63.7435	+0.0826	0.090	0.9284	n.s.	0.013
Model-Free RL (PPO)	100	54.4643	-9.1966	-7.731	0.0000	***	-1.093
Hybrid Tuning RL	100	49.3807	-14.2802	-14.051	0.0000	***	-1.987
Hybrid AS+RL (Proposed)	100	66.2980	+2.6371	1.881	0.0619	n.s.	0.266

Table B. Risk-Adjusted Performance Metrics

Method	Sharpe	Sortino	Calmar	Omega Ratio	Hit Rate	Profit Factor	Downside Dev	Return Skewness	Excess Kurtosis
Constrained AS (Guéant)	7.054	18.600	47.277	5.795	100.00%	5.795	0.1359	1.530	3.010
Static AS	7.042	18.512	47.146	5.769	100.00%	5.769	0.1364	1.527	3.003
Hybrid Tuning RL	5.006	21.783	50.779	6.734	100.00%	6.734	0.0900	2.883	8.450
Model-Free RL (PPO)	4.557	10.609	19.437	2.611	100.00%	2.611	0.2037	1.501	3.877
Hybrid AS+RL (Proposed)	4.452	6.730	13.581	2.260	100.00%	2.260	0.3908	0.201	2.203

*Table A: Welch two-sample t-test; *p<0.05, **p<0.01, ***p<0.001, n.s. = not significant. Cohen's d: |d|<0.2 small, 0.2–0.5 medium, >0.8 large. Table B: Omega ratio = E[gains]/E[losses]; Sortino uses downside deviation.*

Distribution Analysis & Ranking

Table C. Terminal PnL Distribution Moments

Method	N	Mean	Std	Skewness	Excess Kurtosis	Min	Q5	Q25	Median	Q75	Q95	Max
Hybrid AS+RL (Proposed)	100	66.2980	12.3620	-0.059	0.478	30.0261	46.8060	59.0244	65.4973	73.6185	90.2554	94.6033
Constrained AS (Guéant)	100	63.7435	6.4438	0.179	-0.113	51.0979	52.7200	59.4144	63.9650	67.9400	74.1576	82.7591
Static AS	100	63.6609	6.4637	0.168	-0.121	50.9032	52.6176	59.3857	63.8452	67.9400	74.1561	82.5848
Model-Free RL (PPO)	100	54.4643	9.9148	-0.048	1.673	25.6615	40.1186	48.6116	54.0400	60.8204	67.8173	90.9513
Hybrid Tuning RL	100	49.3807	7.7766	-0.358	0.101	25.0931	35.7552	45.4552	49.4758	54.6247	61.5795	65.8081

Table D. Per-Metric Rank (1 = best in column)

Method	Mean PnL	Sharpe	Sortino	Calmar	Hit Rate	Profit Factor	Fill Rate	Max DD	CVaR 5%	Mean |Inv|	Std Inv	Inv Tail	Time@Limit	Downside Dev	Avg Rank
Hybrid Tuning RL	5	3	1	1	4	1	5	1	1	1	1	1	1	1	1.93
Constrained AS (Guéant)	2	1	2	2	2	2	3	2	2	3	3	3	3	2	2.29
Static AS	3	2	3	3	1	3	2	3	3	2	4	2	2	3	2.57
Model-Free RL (PPO)	4	4	4	4	3	4	4	4	4	4	2	4	4	4	3.79
Hybrid AS+RL (Proposed)	1	5	5	5	5	5	1	5	5	5	5	5	5	5	4.43

Table E. CMDP Constraint Satisfaction Rate

Method	DD bound	Mean MaxDD	DD satisfied (%)	DD slack (mean)	CVaR bound	Mean CVaR 5%	CVaR satisfied (%)	CVaR slack (mean)	Both satisfied (%)
Static AS	5.0	1.3503	100.0%	+3.6497	10.0	0.3293	100.0%	+9.6707	100.0%
Constrained AS (Guéant)	5.0	1.3483	100.0%	+3.6517	10.0	0.3282	100.0%	+9.6718	100.0%
Model-Free RL (PPO)	5.0	2.8021	93.0%	+2.1979	10.0	0.6035	100.0%	+9.3965	93.0%
Hybrid Tuning RL	5.0	0.9725	100.0%	+4.0275	10.0	0.2140	100.0%	+9.7860	100.0%
Hybrid AS+RL (Proposed)	5.0	4.8815	60.0%	+0.1185	10.0	1.0172	100.0%	+8.9828	60.0%

Table C: distributional statistics of terminal PnL across evaluation episodes. Table D: rank 1 = best; lower Avg Rank = more consistently strong. Table E: fraction of episodes where the realised constraint metric stays within the CMDP training bound; slack > 0 means the constraint is comfortably satisfied.